Discovery contrast

862 **Layer 1 Programme definition** genes in programme

FDR<0.05, |logFC| ≥ 0.3, endometrial expression gate; 608 up / 254 down

preregistered, hashed, and carried unchanged into Layers 3 and 4

Risk GWAS + cis-eQTL

3 / 503 **Layer 2 Colocalisation** two at 17q21.32

Gene-locus tests across 19 loci and five tissues; PP.H4 > 0.8 threshold

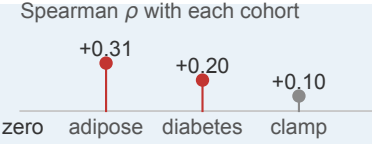
73 of 95 locus-tissue cells testable; one signal survives a change of dataset

(from Layer 1)

+0.31 **Layer 3 Bulk proxy validity**

Rank correlation with three independent cohorts

adipose strong, diabetes intermediate, clamp near zero



(from Layer 1)

−3.8 **Layer 4 Single-cell projection**

11 samples, 141 316 cells; tumour vs normal epithelium, sample-level test

against 100 matched random modules

Finding

One shared causal signal at 17q21.32, in liver (SKAP1, PP.H4 = 0.93) and muscle (0.90); the 57-variant credible set identifies a signal, not a variant; a third signal at 15q15.2 did not survive

Boundary

Uterine eQTL is myometrium, not epithelium; clamp cohort 51 vs 18 limits precision